# NEET UG 2027: Physics Formula Compendium

## Module 01: Kinematics, Dynamics, Work-Energy & Collisions

**Prepared, Compiled & Authored by Satya Prakash** | **Subject:** Physics (Units 2, 3 & 4)

Strict NCERT & NMC Alignment | Complete Mathematical Variations, Use-Cases & Traps

# 1 Kinematics (Motion in a Straight Line & Plane)

## 1.1 Rectilinear Motion under Uniform Acceleration

**Master Kinematic Equations (Constant Acceleration $a = \text{const}$)**

$$v = u + at \tag{1}$$

$$s = ut + \frac{1}{2}at^2 \tag{2}$$

$$v^2 = u^2 + 2as \tag{3}$$

$$s_n = u + \frac{a}{2}(2n - 1) \quad \text{(Displacement in the } n\text{-th second)} \tag{4}$$

$$s = \left(\frac{u + v}{2}\right) t \tag{5}$$

**Parameter Specification:**

- $u$: Initial velocity vector at $t = 0$ [$\mathrm{m\,s^{-1}}$]
- $v$: Instantaneous velocity vector at time $t$ [$\mathrm{m\,s^{-1}}$]
- $a$: Constant acceleration vector [$\mathrm{m\,s^{-2}}$]
- $s$: Net displacement vector from origin [m]
- $s_n$: Displacement accumulated strictly between $t = (n-1)$ and $t = n$ [m]

**Applicability Conditions & Shortcuts**

- **Criterion of Validity:** Strictly applicable **only** when acceleration $a$ is constant in both magnitude and direction. If $a = f(t)$ or $a = f(x)$, kinematic equations fail; one must integrate:

$$v = \int a\,dt, \quad s = \int v\,dt, \quad \int v\,dv = \int a\,dx$$

- **Galileo's Law of Odd Numbers:** For a body starting from rest ($u = 0$) under constant acceleration, the distances traversed in equal successive intervals of time $\Delta t$ follow the ratio:

$$s_1 : s_2 : s_3 : \cdots : s_n = 1 : 3 : 5 : \cdots : (2n - 1)$$

- **Stopping Distance & Stopping Time:**

$$d_{\text{stop}} = \frac{u^2}{2a} \implies d_{\text{stop}} \propto u^2, \qquad t_{\text{stop}} = \frac{u}{a} \implies t_{\text{stop}} \propto u$$

**NEET Traps: Distance vs. Displacement & Turning Points**

- **Trap 1 (Distance vs Displacement):** The formula $s = ut + \frac{1}{2}at^2$ calculates **displacement**, NOT total distance traveled. If velocity changes sign ($v = 0$ at some $t_{\text{turn}} < t_{\text{total}}$), distance

must be calculated piecewise:

$$\text{Distance} = |s(0 \to t_{\text{turn}})| + |s(t_{\text{turn}} \to t_{\text{total}})|$$

- **Trap 2 ($n$-th Second Dimensionality):** In $s_n = u + \frac{a}{2}(2n-1)$, $u$ is multiplied by an implicit (1 s), preserving dimensional homogeneity ($[L] = [LT^{-1} \cdot T] = [L]$). Do not drop signs for $u$ and $a$.

## 1.2 Projectile Motion in a Uniform Gravitational Field

**Oblique Projectile Motion (Launch angle $\theta$ with horizontal)**

$$T = \frac{2u\sin\theta}{g} = \frac{2u_y}{g} \quad \text{[Total Time of Flight]} \tag{6}$$

$$H_{\max} = \frac{u^2\sin^2\theta}{2g} = \frac{u_y^2}{2g} \quad \text{[Maximum Height]} \tag{7}$$

$$R = \frac{u^2\sin 2\theta}{g} = \frac{2u_x u_y}{g} \quad \text{[Horizontal Range]} \tag{8}$$

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta} = x\tan\theta\left(1 - \frac{x}{R}\right) \quad \text{[Trajectory Equation]} \tag{9}$$

**High-Yield Variations & Relations**

- **Relation Between $H_{\max}$ and $R$:**

$$\frac{H_{\max}}{R} = \frac{\tan\theta}{4} \implies R\tan\theta = 4H_{\max}$$

  When launch angle $\theta = 45°$, $R_{\max} = 4H_{\max}$.
- **Complementary Angles of Projection:** For identical launch speed $u$, angles $\theta$ and $(90° - \theta)$ yield identical horizontal ranges ($R_1 = R_2$). The corresponding flight times $T_1, T_2$ and heights $H_1, H_2$ satisfy:

$$R = \frac{1}{2}gT_1T_2 = 4\sqrt{H_1H_2}$$

- **Velocity at Highest Point:** $v_{\text{top}} = u_x = u\cos\theta$ (purely horizontal, acceleration $g$ is perpendicular: $\mathbf{a} \perp \mathbf{v}$).
- **Radius of Curvature:** At apex: $\rho_{\text{apex}} = \frac{u^2\cos^2\theta}{g}$; At launch: $\rho_{\text{launch}} = \frac{u^2}{g\cos\theta}$.

**Common Projectile Traps**

- **Kinetic Energy at Apex:** If initial KE is $K_0 = \frac{1}{2}mu^2$, KE at highest point is **not zero**; it is $K_{\text{apex}} = K_0\cos^2\theta$. For $\theta = 60°$, $K_{\text{apex}} = \frac{1}{4}K_0$.
- **Angle Given with Vertical:** If problem states "projected at angle $\alpha$ to the vertical", replace $\theta$ with $(90° - \alpha)$. Always double check the reference axis!

# 2 Laws of Motion & Friction

## 2.1 Newton's Second Law, Momentum & Impulse

**Momentum, Force & Impulse Formulations**

$$\mathbf{F}_{\text{net}} = \frac{d\mathbf{p}}{dt} = m\frac{d\mathbf{v}}{dt} + \mathbf{v}_{\text{rel}}\frac{dm}{dt} \tag{10}$$

$$\mathbf{J} = \int_{t_1}^{t_2} \mathbf{F}\, dt = \Delta\mathbf{p} = \mathbf{p}_2 - \mathbf{p}_1 \quad \text{[Impulse-Momentum Theorem]} \tag{11}$$

**Variable Mass System (Rocket Propulsion):**

$$v = u + v_{\text{rel}} \ln\left(\frac{m_0}{m}\right) - gt, \qquad F_{\text{thrust}} = v_{\text{rel}}\left(-\frac{dm}{dt}\right) \tag{12}$$

## 2.2 Friction Mechanics

**Static, Limiting & Kinetic Friction Laws**

$$f_s \leq f_{s,\max} = \mu_s N \quad \text{[Static Friction: Self-Adjusting, } 0 \leq f_s \leq \mu_s N\text{]} \tag{13}$$

$$f_k = \mu_k N \quad \text{[Kinetic Friction: Constant for given surfaces, } \mu_k \leq \mu_s\text{]} \tag{14}$$

$$\tan\lambda = \mu_s \quad \text{[Angle of Friction } \lambda\text{]} \tag{15}$$

$$\theta_r = \lambda = \tan^{-1}(\mu_s) \quad \text{[Angle of Repose } \theta_r\text{]} \tag{16}$$

**Motion on an Inclined Plane of Inclination $\theta$**

- If $\theta \leq \theta_r$: Block remains at rest on incline; static friction $f_s = mg\sin\theta$.
- If $\theta > \theta_r$: Block accelerates downward:

$$a_{\text{down}} = g(\sin\theta - \mu_k\cos\theta)$$

- If projected upward along incline:

$$a_{\text{up}} = g(\sin\theta + \mu_k\cos\theta) \implies a_{\text{up}} > a_{\text{down}} \implies t_{\text{up}} < t_{\text{down}}$$

- **Ratio of Times ($n$-times longer down a rough plane vs smooth plane):**

$$\mu = \tan\theta\left(1 - \frac{1}{n^2}\right)$$

**Friction Calculation Traps**

- **Normal Reaction Modification:** $N \neq mg$ in general! On incline: $N = mg\cos\theta$. When pulling at angle $\alpha$: $N = mg - F\sin\alpha$. When pushing at angle $\alpha$: $N = mg + F\sin\alpha$. (Hence pulling is always easier than pushing).
- **Static Friction is NOT Always $\mu_s N$:** $\mu_s N$ is the **maximum limit** (impending motion). If applied force $F_{\text{applied}} < \mu_s N$, the friction force is exactly equal to $F_{\text{applied}}$, NOT $\mu_s N$.

## 2.3 Circular Dynamics & Highway Banking

**Vehicle Dynamics on Circular Tracks**

$$v_{\text{max,flat}} = \sqrt{\mu_s g R} \quad \text{[Unbanked Flat Road]} \tag{17}$$

$$v_{\text{opt}} = \sqrt{gR\tan\theta} \quad \text{[Banked Road, Zero Friction Needed]} \tag{18}$$

$$v_{\text{max,banked}} = \sqrt{gR\left(\frac{\mu_s + \tan\theta}{1 - \mu_s\tan\theta}\right)} \quad \text{[Banked Road with Friction, Upper Speed Limit]} \tag{19}$$

$$v_{\text{min,banked}} = \sqrt{gR\left(\frac{\tan\theta - \mu_s}{1 + \mu_s\tan\theta}\right)} \quad \text{[Banked Road with Friction, Lower Speed Limit]} \tag{20}$$

# 3 Work, Energy, Power & Collisions

## 3.1 Work-Energy Theorem & Potential Energy

**Fundamental Work & Potential Relations**

$$W_{\text{net}} = \Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \quad \text{[Universal Work-Energy Theorem]} \tag{21}$$

$$W_{\text{cons}} = -\Delta U = -(U_f - U_i) \quad \text{[Conservative Force Definition]} \tag{22}$$

$$\mathbf{F} = -\boldsymbol{\nabla} U = -\left(\frac{\partial U}{\partial x}\hat{\mathbf{i}} + \frac{\partial U}{\partial y}\hat{\mathbf{j}} + \frac{\partial U}{\partial z}\hat{\mathbf{k}}\right) \quad \text{[Force-Potential Gradient Relation]} \tag{23}$$

**Spring Elastic Potential Energy:**

$$U_{\text{spring}} = \frac{1}{2}kx^2, \qquad W_{\text{spring}} = -\frac{1}{2}k(x_f^2 - x_i^2) \tag{24}$$

**Equilibrium Criteria from Potential Function $U(x)$**

- **Equilibrium Condition:** $\frac{dU}{dx} = 0$ (Force $F = 0$).
- **Stable Equilibrium:** $\frac{d^2U}{dx^2} > 0$ (Potential energy is minimum; restoring force acts on perturbation; SHM possible).
- **Unstable Equilibrium:** $\frac{d^2U}{dx^2} < 0$ (Potential energy is maximum; destabilizing force acts on perturbation).
- **Neutral Equilibrium:** $\frac{d^2U}{dx^2} = 0$ (Potential energy is locally constant).

## 3.2 Vertical Circular Motion (Mass $m$ on Inextensible String Length $L$)

**Critical Velocity & Tension Conditions**

$$v_{\text{bottom,min}} = \sqrt{5gL} \quad \text{[Condition to complete full circle]} \tag{25}$$

$$v_{\text{top,min}} = \sqrt{gL} \quad \text{[Velocity at top for non-slack string } (T_{\text{top}} \geq 0)\text{]} \tag{26}$$

$$v_{\text{horizontal}} = \sqrt{3gL} \quad \text{[Velocity when string is horizontal]} \tag{27}$$

$$T_{\text{bottom}} - T_{\text{top}} = 6mg \quad \text{[Tension difference is independent of launch speed!]} \tag{28}$$

**String vs Light Rigid Rod in Vertical Circle**

- **String (can slack):** Minimum velocity at bottom to complete circle is $\sqrt{5gL}$; minimum velocity at apex is $\sqrt{gL}$.
- **Light Rigid Rod (cannot slack):** Velocity at highest point can drop to **zero** ($v_{\text{top}} = 0$). Hence minimum velocity at bottom is:

$$v_{\text{bottom,min}} = \sqrt{4gL} = 2\sqrt{gL}$$

Never confuse the string case with the rigid rod / sphere on track case!

## 3.3 1D & 2D Collisions

### Coefficient of Restitution & Collision Velocities

$$e = \frac{v_{\text{sep}}}{v_{\text{app}}} = \frac{v_2 - v_1}{u_1 - u_2} \quad (0 \le e \le 1) \tag{29}$$

**1D Head-On Collision Velocities ($m_1$ moving with $u_1$, $m_2$ moving with $u_2$):**

$$v_1 = \left(\frac{m_1 - em_2}{m_1 + m_2}\right) u_1 + \frac{(1+e)m_2}{m_1 + m_2} u_2 \tag{30}$$

$$v_2 = \frac{(1+e)m_1}{m_1 + m_2} u_1 + \left(\frac{m_2 - em_1}{m_1 + m_2}\right) u_2 \tag{31}$$

**Loss of Kinetic Energy in Inelastic Collision ($\Delta K$):**

$$\Delta K_{\text{loss}} = \frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)(u_1 - u_2)^2(1 - e^2) = \frac{1}{2}\mu v_{\text{rel}}^2(1 - e^2) \tag{32}$$

### Special Elastic Collision Limits ($e = 1$)

- **Equal Masses ($m_1 = m_2$):** Velocities simply interchange after collision:

$$v_1 = u_2, \qquad v_2 = u_1$$

- **Heavy Target at Rest ($m_2 \gg m_1, u_2 = 0$):** $v_1 \approx -u_1$ (rebounds with same speed), $v_2 \approx 0$.
- **Heavy Projectile against Light Target ($m_1 \gg m_2, u_2 = 0$):** $v_1 \approx u_1$, $v_2 \approx 2u_1$ (target launched at twice projectile speed).
- **Rebound from Floor (Floor $m_2 \to \infty, u_2 = 0$):**

$$v_n = e^n v_0, \qquad h_n = e^{2n} h_0, \qquad t_{\text{total}} = t_0\left(\frac{1+e}{1-e}\right), \qquad H_{\text{total}} = h_0\left(\frac{1+e^2}{1-e^2}\right)$$

**End of Module 01 Formula Handbook**

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